

Problem determination and troubleshooting

How to perform problem determination actions on the SR680a V3

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Problem determination and troubleshooting overview

Perform the following actions to determine the cause of problems on the SR680a V3

- Check the system health status on the XCC2 dashboard or the operation panel at the front of the system
- Check the system event log in XCC2
- Check the event log in UEFI
- Check the LEDs on the system
- Integrated LCD diagnostics panel

For more information about how to use XCC2, UEFI, or OneCLI to monitor system status and collect logs, refer to the following courses:

- [ES51757B – Introducing ThinkSystem tools](#)
- [ES52374 – ThinkSystem tools for the ThinkSystem V3 platform](#)
- [ES41759C – ThinkSystem problem determination](#)

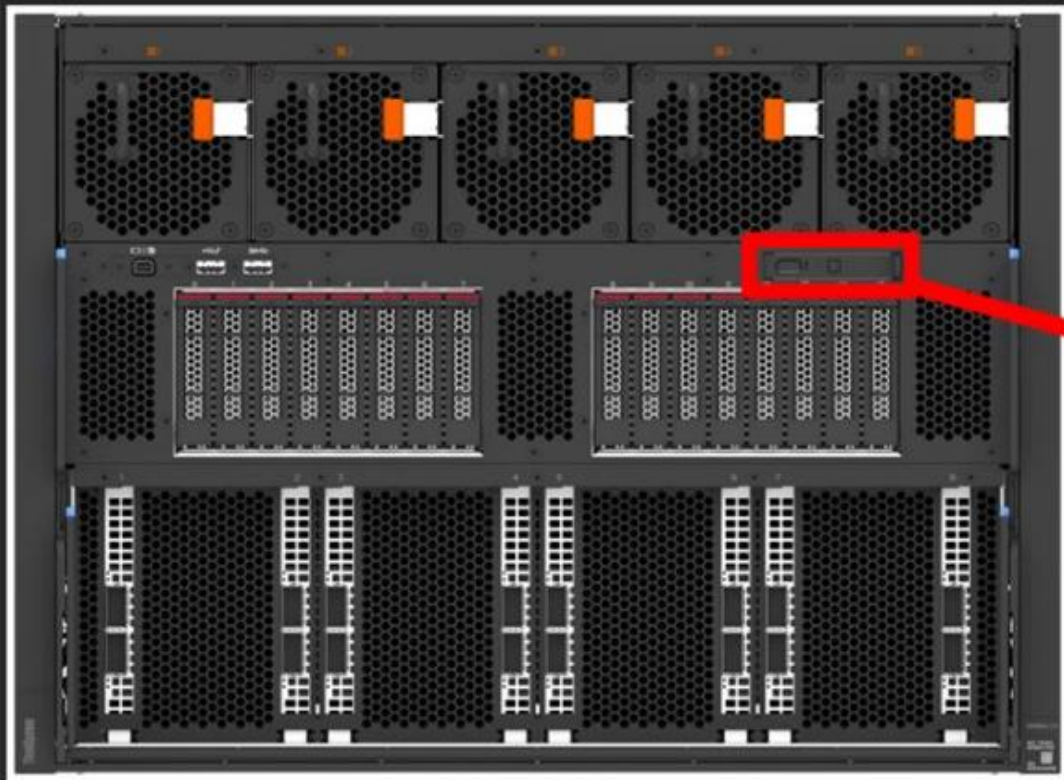
Note: The SR680a V3 does not support the optional external diagnostic panel.

LED descriptions

- Use the LEDs on the front drives, front operator panel, and the rear side of the server for hardware status monitoring and problem determination.
- The front and rear fans do not have status LEDs.
- The system board assembly does not support the internal LED light path feature. As the SR680a V3 does not have a top cover, you will not be able to see the system board assembly unless you power off the system and remove the system board tray from the chassis.
- Click [HERE](#) to see the location of the SR680a V3 front LCD diagnostic panel.
- For more information about the SR680a V3 LEDs, refer to the *Server components* section of the *ThinkSystem SR680a V3 User Guide* on [Lenovo Docs](#).

LED descriptions

SR680a V3 front LCD diagnostic panel



A demo video describing how the LCD diagnostic panel is used is available on the course landing page.

GPU problem determination

Use the following tools to monitor or collect service data for GPU problem escalation:

- Use XCC to check the GPU inventory information
- If necessary, use the following command line tools in the OS to collect service data
 - Linux commands
 - NVIDIA / AMD driver bundle SMI (system management interface) CLI tool

Note:

- If a GPU problem occurs that might require a GPU or GPU baseboard replacement, escalate to the PE level of support for further problem diagnosis.
- The GPU connector and GPU slot are fragile. To avoid damaging either component, do not move a potentially failed GPU to another slot for problem determination.

Monitoring GPU status with XCC

GPU inventory and firmware information is available in XCC. Servicers can also find GPU sensor information in the XCC logs.

Click the following links to see screenshots:

- [H100 GPU inventory list in XCC](#)
- [H100 GPU details in XCC](#)
- [GPU inventory examples in the XCC FFDC mini log](#)
- [GPU error message in XCC](#)

Monitoring GPU status with XCC

H100 GPU inventory list in XCC



Slot	Device Name	Device Type	Card Interface	FRU Number
OnBoard	Onboard VGA		OnBoard	N/A
11	Broadcom 57414 10/25GbE SFP28 2-port OCP Ethernet Adapter		OCP	03GX532
	Broadcom 57414 10/25GbE SFP28 2-port OCP Ethernet Adapter 00:26:00:00	Ethernet	OCP	
	Broadcom 57414 10/25GbE SFP28 2-port OCP Ethernet Adapter 00:28:00:01	Ethernet	OCP	
17	H100 80GB HBM3		PCI-E	N/A
18	H100 80GB HBM3		PCI-E	N/A
19	H100 80GB HBM3		PCI-E	N/A
20	H100 80GB HBM3		PCI-E	N/A
21	H100 80GB HBM3		PCI-E	N/A
22	H100 80GB HBM3		PCI-E	N/A
23	H100 80GB HBM3		PCI-E	N/A
24	H100 80GB HBM3		PCI-E	N/A
33	ThinkSystem M.2 NVMe 2-Bay RAID Enablement Kit	NVMe Controller	PCI-E x4	03LD510

The GPUs are listed on the XCC **Inventory** page (PCIe slots 17 to 24).

Monitoring GPU status with XCC

H100 GPU details in XCC

PCI Adapters

Slot	Device Name	Device Type	Card Interface	FRU Number
OnBoard	ASPEED AST2600 VGA		OnBoard	N/A
17	H100 80GB HBM3		PCI-E	N/A

▼ Asset Summary

UUID	N/A	Manufacturer	NVIDIA
Serial Number		Model	N/A
FRU Number	N/A	Part Number	2330-885-A1
Package	Module/Card	Max Data Width	16

▼ PCI Summary

Segment Number	0	Bus Number	24
Device Number	0	Function Number	0
Vendor ID	0x10DE	Device ID	0x2330
Sub Vendor ID	0x10DE	Sub Device ID	0x16C1
Slot Designation	Slot 17	Support Hot Plug	No
Revision ID	0x00A1		

▼ Firmware

		Firmware				
Classification	Description	Name	Manufacturer	Release Date	Software ID	Version
Software Bundle	Firmware package containing controller firmware, UEFI Driver, and other sub components.	Software Bundle	NVIDIA		10DE16C1	96.00.89.00.01

The SR680a V3 supports an all-in-one software bundle package for all eight GPUs and the GPU baseboard.

Monitoring GPU status with XCC

GPU inventory examples in the XCC FFDC mini log



Adapter

slotNo	adapterName	connectorLabel	oobSupported	location	functions
18	M1300X	PCI-E	1	1	funType: 16 oobType: 1 generic_segNo: 0 generic_busNo: 133 generic_devNo: 0 generic_funcNo: 0 generic_vendorId: 4098 generic_devId: 29857 generic_revId: 0 generic_subVendor: 4098 generic_subDevId: 29857 generic_slotNo: 18 generic_slotDesignation: Slot 18 generic_hotplug: 0 vpd_uuid: N/A vpd_cardName: M1300X vpd_manufacturer: AMD vpd_serialNo: 692402006379 vpd_partNo: 100-300000065H vpd_model: N/A vpd_cardSKU: N/A misc_fodUID: N/A misc_maxDataWidth: 16 misc_package: Module/Card firmwares: firmwareName: Software Bundle versionStr: 00.24.06.12 description: Firmware package containing controller firmware, UEFI Driver, and other sub components releaseDate: classification: Software Bundle manufacturer: AMD softwareID: 100274A1
19	M1300X	PCI-E	1	1	funType: 16 oobType: 1 generic_segNo: 0 generic_busNo: 229 generic_devNo: 0 generic_funcNo: 0 generic_vendorId: 4098 generic_devId: 29857 generic_revId: 0 generic_subVendor: 4098 generic_subDevId: 29857 generic_slotNo: 19 generic_slotDesignation: Slot 19 generic_hotplug: 0 vpd_uuid: N/A vpd_cardName: M1300X vpd_manufacturer: AMD vpd_serialNo: 692402006445

IPMI Sensor

sensor name	value	lower critical threshold	lower noncritical threshold	lower nonrecoverable threshold	status	unit
PCIe 17	0x0	na	na	na	0x0480	discrete
PCIe 17 OverTemp	0x0	na	na	na	0x0180	discrete
PCIe 17 Temp	45.000	na	na	na	ok	degrees C
PCIe 18	0x0	na	na	na	0x0480	discrete
PCIe 18 OverTemp	0x0	na	na	na	0x0180	discrete
PCIe 18 Temp	53.000	na	na	na	ok	degrees C
PCIe 19	0x0	na	na	na	0x0480	discrete
PCIe 19 OverTemp	0x0	na	na	na	0x0180	discrete
PCIe 19 Temp	50.000	na	na	na	ok	degrees C
PCIe 20	0x0	na	na	na	0x0480	discrete
PCIe 20 OverTemp	0x0	na	na	na	0x0180	discrete
PCIe 20 Temp	46.000	na	na	na	ok	degrees C
PCIe 21	0x0	na	na	na	0x0480	discrete
PCIe 21 OverTemp	0x0	na	na	na	0x0180	discrete
PCIe 21 Temp	47.000	na	na	na	ok	degrees C
PCIe 22	0x0	na	na	na	0x0480	discrete
PCIe 22 OverTemp	0x0	na	na	na	0x0180	discrete
PCIe 22 Temp	49.000	na	na	na	ok	degrees C
PCIe 23	0x0	na	na	na	0x0480	discrete
PCIe 23 OverTemp	0x0	na	na	na	0x0180	discrete
PCIe 23 Temp	49.000	na	na	na	ok	degrees C
PCIe 24	0x0	na	na	na	0x0480	discrete
PCIe 24 OverTemp	0x0	na	na	na	0x0180	discrete
PCIe 24 Temp	46.000	na	na	na	ok	degrees C

The GPUs are listed in the **Adapter** section (slots 17 to 24)

The **IPMI Sensor** section shows GPU temperature status information (PCIe 17 to 24)

Monitoring GPU status with XCC

GPU error messages in XCC



2

- ❌ PCI Fault in slot 22 on system ThinkSystem SR685a V3 System Board.
FQXSPIO0015M FRU: June 3, 2024 1:01:34 PM
- ❌ PCI An Uncorrectable PCIe Error has Occurred at Bus 02 Device 00 Function 00. The Vendor ID for the device is 1000 and the Device ID is C030. The physical slot number is 22.
FQXSPIO0010M FRU: June 3, 2024 1:01:28 PM

XCC index page

The error messages for PCIe slots 17 to 24 indicate a possible GPU error incident.

881	❌	System	FQXSPIO0015M	Fault in slot 22 on system ThinkSystem SR685a V3 System Board.
882	❌	System	FQXSPIO0006N	A software NMI has occurred on system ThinkSystem SR685a V3 System Board.
883	❌	System	FQXSPIO0010M	An Uncorrectable PCIe Error has Occurred at Bus 02 Device 00 Function 00. The Ve...

XCC event log page

Note: These screenshots are from an SR685a, but the SR680a has the same GPU error messages in XCC.

Monitoring GPU status with Linux commands

In a Linux environment, use the following commands to collect GPU-related information or the system log and then export the command line output as a txt file for further analysis:

- Collecting GPU information: `dmesg | grep <amd or nvidia> > <folder/filename>`

Example: `dmesg | grep amd > /tmp/dmesg.txt`

```
root@monaco:/# dmesg | grep amd > /tmp/dmesg.txt
```



```
root@monaco:/# cat /tmp/dmesg.txt
[ 0.000000] Linux version 5.15.0-108-generic (build@lcy02-amd64-007) (gcc (Ubuntu 11.4.0-1ubuntu1-22.0
4) 11.4.0, GNU ld (GNU Binutils for Ubuntu) 2.38) #115-Ubuntu SMP Mon Apr 15 09:52:04 UTC 2024 (Ubuntu 5.1
5.0-108.115-generic 5.15.108)
[ 1.918209] perf/amd_iommu: Detected AMD IOMMU #0 (2 banks, 4 counters/bank).
[ 1.918270] perf/amd_iommu: Detected AMD IOMMU #1 (2 banks, 4 counters/bank).
[ 1.918291] perf/amd_iommu: Detected AMD IOMMU #2 (2 banks, 4 counters/bank).
[ 1.918316] perf/amd_iommu: Detected AMD IOMMU #3 (2 banks, 4 counters/bank).
[ 1.918338] perf/amd_iommu: Detected AMD IOMMU #4 (2 banks, 4 counters/bank).
[ 1.918359] perf/amd_iommu: Detected AMD IOMMU #5 (2 banks, 4 counters/bank).
[ 1.918383] perf/amd_iommu: Detected AMD IOMMU #6 (2 banks, 4 counters/bank).
[ 1.918408] perf/amd_iommu: Detected AMD IOMMU #7 (2 banks, 4 counters/bank).
[ 3.717962] amdcl: loading out-of-tree module taints kernel.
[ 3.717962] amdcl: loading out-of-tree module taints kernel.
[ 3.718017] amdcl: module verification failed: signature and/or required key missing - tainting kernel
[ 4.055041] [drm] amdgpu kernel modesetting enabled.
[ 4.056679] [drm] amdgpu version: 6.7.0
[ 4.063171] amdgpu: Virtual CRAT table created for CPU
[ 4.072045] amdgpu: Topology: Add CPU node
[ 4.282500] amdgpu: PeerDirect support was initialized successfully
[ 4.350903] amdgpu 0000:45:00.0: amdgpu: Fetched VBIOS from ROM BAR
[ 4.351626] amdgpu: ATOM BIOS: 113-M13SR10V-001
[ 4.352467] amdgpu 0000:45:00.0: [drm:vcn_v4_0_3_early_init [amdgpu]] VCN decode is enabled in VM mode
[ 4.353928] amdgpu 0000:45:00.0: [drm:jpeg_v4_0_3_early_init [amdgpu]] JPEG decode is enabled in VM mod
```

- Collecting error messages in the system log folder:

`dmesg | grep -i error /var/log/<system log folder name depends on the OS> > <output folder/filename>`

Example: `dmesg | grep -i error /var/log/syslog > /tmp/amdsyslog.txt`

Monitoring GPU status with SMI commands

System Management Interface (SMI) is a command line utility bundle for GPU drivers. SMI commands can be used to manage and monitor GPU devices. NVIDIA and AMD GPUs support their own version of SMI.

Click the following links to see examples of SMI command outputs:

- [nvidia-smi](#) (to display NVIDIA GPUs installed in the system)
- [nvidia-smi -L](#) (to display NVIDIA GPUs installed in the system with UUID)
- [nvidia-smi -q --id=1 -f <output file name>](#) (to export NVIDIA GPU inventory information)
- [nvidia-smi pci --getErrorCounters](#) (to display NVIDIA GPUs error counters data)
- [rocm-smi](#) (to display AMD GPUs installed in the system)
- [rocm-smi --showrasinfo](#) (to display AMD GPU error counter data)
- [rocm-smi --showhw](#) (to display AMD GPU hardware details)
- [rocm-smi -a](#) (to display more AMD GPU status information)

For more information about NVIDIA and AMD SMI information, refer to the following websites:

- [NVIDIA SMI](#)
- [AMD ROCm SMI](#)

```
nonzagmonza:~$ nvidia-smi
Sat Jun 15 15:06:31 2024
```

NVIDIA-SMI 550.90.07 Driver Version: 550.90.07 CUDA Version: 12.4							
GPU	Name	Perf	Persistence-M	Bus-Id	Disp.A	Volatile Uncorr.	ECC
Fan	Temp		Pwr:Usage/Cap		Memory-Usage	GPU-Util	Compute M. MIG M.
0	NVIDIA H100 80GB HBM3	Off	00000000:1B:00.0 Off		0		
N/A	43C P0	77W / 700W	1MiB / 81559MiB		0%	Default Disabled	
1	NVIDIA H100 80GB HBM3	Off	00000000:29:00.0 Off		0		
N/A	41C P0	79W / 700W	1MiB / 81559MiB		0%	Default Disabled	
2	NVIDIA H100 80GB HBM3	Off	00000000:3A:00.0 Off		0		
N/A	44C P0	76W / 700W	1MiB / 81559MiB		0%	Default Disabled	
3	NVIDIA H100 80GB HBM3	Off	00000000:5C:00.0 Off		0		
N/A	45C P0	73W / 700W	1MiB / 81559MiB		0%	Default Disabled	
4	NVIDIA H100 80GB HBM3	Off	00000000:9A:00.0 Off		0		
N/A	42C P0	74W / 700W	1MiB / 81559MiB		0%	Default Disabled	
5	NVIDIA H100 80GB HBM3	Off	00000000:AA:00.0 Off		0		
N/A	40C P0	75W / 700W	1MiB / 81559MiB		0%	Default Disabled	
6	NVIDIA H100 80GB HBM3	Off	00000000:BA:00.0 Off		0		
N/A	40C P0	75W / 700W	1MiB / 81559MiB		0%	Default Disabled	
7	NVIDIA H100 80GB HBM3	Off	00000000:CA:00.0 Off		0		
N/A	42C P0	78W / 700W	1MiB / 81559MiB		0%	Default Disabled	

```
Processes:
```

GPU	GI	CI	PID	Type	Process name	GPU Memory Usage
ID	ID	ID				
No running processes found						

```
nonzagmonza:~$
```


Monitoring GPU status with SMI commands

```
nvidia-smi -L
```



```
monza@monza:~$ nvidia-smi -L
GPU 0: NVIDIA H100 80GB HBM3 (UUID: GPU-6e0a65fb-718e-5b02-59f6-8299cf79d5ff)
GPU 1: NVIDIA H100 80GB HBM3 (UUID: GPU-1feb659e-68d7-989b-f7a5-ee58dd99022e)
GPU 2: NVIDIA H100 80GB HBM3 (UUID: GPU-0896702e-cdb2-6600-b0a7-8ccc184e6d1d)
GPU 3: NVIDIA H100 80GB HBM3 (UUID: GPU-0963c80d-fb0a-136e-895a-243459c6023f)
GPU 4: NVIDIA H100 80GB HBM3 (UUID: GPU-e30aaa97-7c92-5395-899f-fb09ab23b9e2)
GPU 5: NVIDIA H100 80GB HBM3 (UUID: GPU-94ab9e89-76fb-7428-df61-023cf4b7751e)
GPU 6: NVIDIA H100 80GB HBM3 (UUID: GPU-6fc98cc6-d0d4-a04b-16b1-1e629800d849)
GPU 7: NVIDIA H100 80GB HBM3 (UUID: GPU-4cf011b1-5de1-d8d6-a26a-b48961e1d5c8)
monza@monza:~$
```

Monitoring GPU status with SMI commands

```
nvidia-smi -q --id=1 -f <output file name>
```



```
nvidia-smi -q --id=1 -f /tmp/queryoam1.txt
```



```
*****NVSMI LOG*****
Timestamp                : Sat Jun 15 15:12:42 2024
Driver Version           : 550.90.07
CUDA Version             : 12.4

Attached GPUs            : 8
GPU 00000000:29:00.0
  Product Name           : NVIDIA H100 80GB HBM3
  Product Brand          : NVIDIA
  Product Architecture    : Hopper
  Display Mode            : Enabled
  Display Active         : Disabled
  Persistence Mode       : Disabled
  Addressing Mode        : None
  MIG Mode
    Current              : Disabled
    Pending              : Disabled
  Accounting Mode        : Disabled
  Accounting Mode Buffer Size : 4000
  Driver Model
    Current              : N/A
    Pending              : N/A
  Serial Number          : 1654123019435
  GPU UUID               : GPU-1feb659e-68d7-989b-f7a5-ec586d99022e
  Minor Number           : 1
  VBIOS Version          : 96.00.89.00.01
  MultiGPU Board         : No
  Board ID               : 0x2900
  Board Part Number      : 692-2G520-0200-000
  GPU Part Number        : 2330-885-A1
  FRU Part Number        : N/A
  Module ID              : 8
  Inforom Version
    Image Version        : G520.0200.00.05
    OEM Object           : 2.1
    ECC Object           : 7.16
    Power Management Object : N/A
  Inforom BBX Object Flush
    Latest Timestamp     : N/A
    Latest Duration      : N/A
  GPU Operation Mode
    Current              : N/A
    Pending              : N/A
  GPU C2C Mode           : Disabled
  GPU Virtualization Mode
    Virtualization Mode  : None
    Host VGPU Mode       : N/A
    GPU Heterogeneous Mode : N/A
  GPU Reset Status
    Reset Required       : No
```


Monitoring GPU status with SMI commands

`nvidia-smi pci --getErrorCounters`



```
monza@monza:~$ nvidia-smi pci --getErrorCounters
GPU 0: NVIDIA H100 80GB HBM3 (UUID: GPU-6e0a65fb-718e-5b02-59f6-8299cf79d5ff)
REPLAY_COUNTER: 0
REPLAY_ROLLOVER_COUNTER: 0
LO_TO_RECOVERY_COUNTER: 5
CORRECTABLE_ERRORS: 0
NAKS_RECEIVED: 0
RECEIVER_ERROR: 0
BAD_TLP: 0
NAKS_SENT: 0
BAD_DLLP: 0
NON_FATAL_ERROR: 0
FATAL_ERROR: 0
UNSUPPORTED_REQ: 0
LCRC_ERROR: 0
LANE_ERROR:
  lane 0: 0
  lane 1: 0
  lane 2: 0
  lane 3: 0
  lane 4: 0
  lane 5: 0
  lane 6: 0
  lane 7: 0
  lane 8: 0
  lane 9: 0
  lane 10: 0
  lane 11: 0
  lane 12: 0
GPU 1: NVIDIA H100 80GB HBM3 (UUID: GPU-1feb659e-68d7-989b-f7a5-ee58dd99022e)
REPLAY_COUNTER: 0
REPLAY_ROLLOVER_COUNTER: 0
LO_TO_RECOVERY_COUNTER: 5
CORRECTABLE_ERRORS: 0
NAKS_RECEIVED: 0
RECEIVER_ERROR: 0
```

The `-pci --getErrorCounters` command displays all the GPU error counters in the system.

```
monza@monza:~$ nvidia-smi pci --getErrorCounters --id=2
GPU 2: NVIDIA H100 80GB HBM3 (UUID: GPU-0896702e-cdb2-6600-b0a7-8ccc184e6d1d)
REPLAY_COUNTER: 0
REPLAY_ROLLOVER_COUNTER: 0
LO_TO_RECOVERY_COUNTER: 5
CORRECTABLE_ERRORS: 0
NAKS_RECEIVED: 0
RECEIVER_ERROR: 0
BAD_TLP: 0
NAKS_SENT: 0
BAD_DLLP: 0
NON_FATAL_ERROR: 0
FATAL_ERROR: 0
UNSUPPORTED_REQ: 0
LCRC_ERROR: 0
LANE_ERROR:
  lane 0: 0
  lane 1: 0
  lane 2: 0
  lane 3: 0
  lane 4: 0
  lane 5: 0
  lane 6: 0
  lane 7: 0
  lane 8: 0
  lane 9: 0
  lane 10: 0
  lane 11: 0
  lane 12: 0
monza@monza:~$
```

Add `--id=<id number>` to display error counters for a specific GPU.

Monitoring GPU status with SMI commands

rocm-smi



```
root@monaco:/# rocm-smi
```

```
===== ROCm System Management Interface =====
===== Concise Info =====
Device [Model : Revision] Temp Power Partitions SCLK MCLK Fan Perf PwrCap VRAM% GPU%
      Name (20 chars) (Junction) (Socket) (Mem, Compute)
=====
0 [0x74a1 : 0x00] 49.0°C 137.0W NPS1, SPX 131Mhz 900Mhz 0% auto 750.0W 0% 0%
  AMD Instinct MI300X
1 [0x74a1 : 0x00] 46.0°C 133.0W NPS1, SPX 131Mhz 900Mhz 0% auto 750.0W 0% 0%
  AMD Instinct MI300X
2 [0x74a1 : 0x00] 53.0°C 137.0W NPS1, SPX 131Mhz 900Mhz 0% auto 750.0W 0% 0%
  AMD Instinct MI300X
3 [0x74a1 : 0x00] 45.0°C 135.0W NPS1, SPX 132Mhz 900Mhz 0% auto 750.0W 0% 0%
  AMD Instinct MI300X
4 [0x74a1 : 0x00] 48.0°C 135.0W NPS1, SPX 131Mhz 900Mhz 0% auto 750.0W 0% 0%
  AMD Instinct MI300X
5 [0x74a1 : 0x00] 46.0°C 133.0W NPS1, SPX 131Mhz 900Mhz 0% auto 750.0W 0% 0%
  AMD Instinct MI300X
6 [0x74a1 : 0x00] 49.0°C 140.0W NPS1, SPX 132Mhz 900Mhz 0% auto 750.0W 0% 0%
  AMD Instinct MI300X
7 [0x74a1 : 0x00] 47.0°C 134.0W NPS1, SPX 131Mhz 900Mhz 0% auto 750.0W 0% 0%
  AMD Instinct MI300X
=====
===== End of ROCm SMI Log =====
```

```
root@monaco:/# |
```


Monitoring GPU status with SMI commands

```
rocm-smi --showrasinfo
```

```
===== ROCm System Management Interface =====
===== RAS Info =====

GPU[0]:      RAS INFO
            Block      Status      Correctable Error      Uncorrectable Error
            UMC         ENABLED         0                      0
            SDMA        ENABLED         0                      0
            GFX         ENABLED         0                      0
            MMHUB       ENABLED         0                      0
            ATHUB       ENABLED
            PCIE_BIF    ENABLED         0                      0
            HDP         ENABLED         0                      0
            XGMI_WAFL   ENABLED         0                      0
            DF          ENABLED
            SMN         ENABLED
            SEM         ENABLED
            MP0         ENABLED
            MP1         ENABLED
            FUSE        ENABLED

-----

GPU[1]:      RAS INFO
            Block      Status      Correctable Error      Uncorrectable Error
            UMC         ENABLED         0                      0
            SDMA        ENABLED         0                      0
            GFX         ENABLED         0                      0
            MMHUB       ENABLED         0                      0
            ATHUB       ENABLED
            PCIE_BIF    ENABLED         0                      0
            HDP         ENABLED         0                      0
            XGMI_WAFL   ENABLED         0                      0
            DF          ENABLED
            SMN         ENABLED
            SEM         ENABLED
            MP0         ENABLED
            MP1         ENABLED
            FUSE        ENABLED

-----

GPU[2]:      RAS INFO
            Block      Status      Correctable Error      Uncorrectable Error
            UMC         ENABLED         0                      0
            SDMA        ENABLED         0                      0
```



Monitoring GPU status with SMI commands

```
rocm-smi --showhw
```



```
root@monaco:/# rocm-smi --showhw
```

```
===== ROCm System Management Interface =====  
===== Concise Hardware Info =====  
GPU  DID  DREV  GFX RAS  SDMA RAS  UMC RAS  VBIOS  BUS  
0    74a1  0     ENABLED  ENABLED  ENABLED  113-MI3SRIOV-001  0000:05:00.0  
1    74a1  0     ENABLED  ENABLED  ENABLED  113-MI3SRIOV-001  0000:25:00.0  
2    74a1  0     ENABLED  ENABLED  ENABLED  113-MI3SRIOV-001  0000:45:00.0  
3    74a1  0     ENABLED  ENABLED  ENABLED  113-MI3SRIOV-001  0000:65:00.0  
4    74a1  0     ENABLED  ENABLED  ENABLED  113-MI3SRIOV-001  0000:85:00.0  
5    74a1  0     ENABLED  ENABLED  ENABLED  113-MI3SRIOV-001  0000:A5:00.0  
6    74a1  0     ENABLED  ENABLED  ENABLED  113-MI3SRIOV-001  0000:C5:00.0  
7    74a1  0     ENABLED  ENABLED  ENABLED  113-MI3SRIOV-001  0000:E5:00.0  
===== End of ROCm SMI Log =====  
root@monaco:/# |
```


Monitoring GPU status with SMI commands

rocm-smi -a



```
===== ROCm System Management Interface =====
===== Version of System Component =====
Driver version: 6.7.0
===== ID =====
GPU[0]      : Device ID: 0x74a1
GPU[0]      : Device Rev: 0x0
GPU[1]      : Device ID: 0x74a1
GPU[1]      : Device Rev: 0x0
GPU[2]      : Device ID: 0x74a1
GPU[2]      : Device Rev: 0x0
GPU[3]      : Device ID: 0x74a1
GPU[3]      : Device Rev: 0x0
GPU[4]      : Device ID: 0x74a1
GPU[4]      : Device Rev: 0x0
GPU[5]      : Device ID: 0x74a1
GPU[5]      : Device Rev: 0x0
GPU[6]      : Device ID: 0x74a1
GPU[6]      : Device Rev: 0x0
GPU[7]      : Device ID: 0x74a1
GPU[7]      : Device Rev: 0x0
===== Unique ID =====
GPU[0]      : Unique ID: 0x42fe8bb12343327a
GPU[1]      : Unique ID: 0x2f805b43b81b9a64
GPU[2]      : Unique ID: 0x5bb9a84de9ed1eb1
GPU[0]      : VBIOS version: 113-MI3SRIOV-001
GPU[7]      : VBIOS version: 113-MI3SRIOV-001
===== Temperature =====
GPU[0]      : Temperature (Sensor junction) (C): 49.0
GPU[0]      : Temperature (Sensor memory) (C): 45.0
GPU[1]      : Temperature (Sensor junction) (C): 45.0
GPU[1]      : Temperature (Sensor memory) (C): 42.0
GPU[2]      : Temperature (Sensor junction) (C): 53.0
GPU[2]      : Temperature (Sensor memory) (C): 48.0
GPU[3]      : Temperature (Sensor junction) (C): 45.0
GPU[3]      : Temperature (Sensor memory) (C): 43.0
GPU[4]      : Temperature (Sensor junction) (C): 48.0
GPU[4]      : Temperature (Sensor memory) (C): 46.0
GPU[5]      : Temperature (Sensor junction) (C): 46.0
GPU[5]      : Temperature (Sensor memory) (C): 43.0
GPU[6]      : Temperature (Sensor junction) (C): 48.0
GPU[6]      : Temperature (Sensor memory) (C): 46.0
GPU[7]      : Temperature (Sensor junction) (C): 46.0
GPU[7]      : Temperature (Sensor memory) (C): 43.0
===== Current clock frequencies =====
GPU[0]      : fclk clock level: 0: (1300Mhz)
GPU[0]      : mclk clock level: 0: (900Mhz)
GPU[0]      : sclk clock level: 1: (141Mhz)
GPU[0]      : socclk clock level: S: (29Mhz)
GPU[1]      : fclk clock level: 0: (1300Mhz)
GPU[1]      : mclk clock level: 0: (900Mhz)
```

The output of `rocm-smi -a` includes details of AMD GPU status, such as temperature or clock frequencies.

NVIDIA GPU and GPU board problem determination

The problem determination steps for NVIDIA GPUs are the same as those used with the previous generation of GPUs. If necessary, collect XCC service data or use the `nvidia-bug-report.sh` command to collect a GPU bug report for problem escalation.

Refer to the following article for complete instructions of how to monitor NVIDIA GPU status and how to collect NVIDIA GPU logs: <https://support.lenovo.com/tw/en/solutions/ht512069>

Health check for GPUs and GPU boards

1. To get the GPU health status, use the following IPMI command:

```
ipmitool -I lanplus -H XCCIPAddress -U USERID -P PASSWORD sdr elist
```

The XCC IP address and the login credentials will vary based on your environment.

If a Linux-based OS is installed on the host system, add the `grep` command to search for GPU information only:

```
ipmitool -I lanplus -H XCCIPAddress -U USERID -P PASSWORD sdr elist | grep GPU
```

Sample output

```
[root@kchen33-flzn8cz ~]# ipmitool sdr elist | grep GPU
GPU Board Power | 8Ch | ok | 21.4 | 240 Watts
GPU Board      | E9h | ok | 11.8 | Transition to OK
GPU CPUs       | EAh | ok | 11.9 | Transition to OK
```

From the output above, `Transition to OK` means the GPU board and the GPU processors are good. If the output shows `Failed`, the system has failed to detect the GPU board or GPU processors.

2. Run the `nvidia-smi` utility to monitor and manage online GPUs. A summary table will be displayed with information about